Trophic model of the outer continental shelf and upper slope demersal community of the Southeastern Brazilian Bight



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REVIZEE



Inventory of living resouces from Brazilian EEZ
Bottom trawl; mid water trawl; traps; “pargueira”, driftnets; longline; Benthic samplers (Van Veen and Box-corer); rectangular drodge

Southeastern Brazilian Bight



227 species of fish in demersal prospection
More abundant 400-600

Outer continental shelf and upper slope (100 – 600 m)
81,685.00 Km^2
Year: 2001

668



Urophycis mystacea
(Gadidae, Gadiformes)

80



Genypterus Brasiliensis
(Ophidiidae, Ophidiiformes)

2579



Synagrops spinosus
(Acropomatidae, Perciforme)

1691



Synagrops bellus
(Acropomatidae, Perciforme)

193 stomachs



Ventrifossa macropogon
(Macrouridae, Gadiformes)



Antigonia capros
(Caproidae, Zeiforme)

404

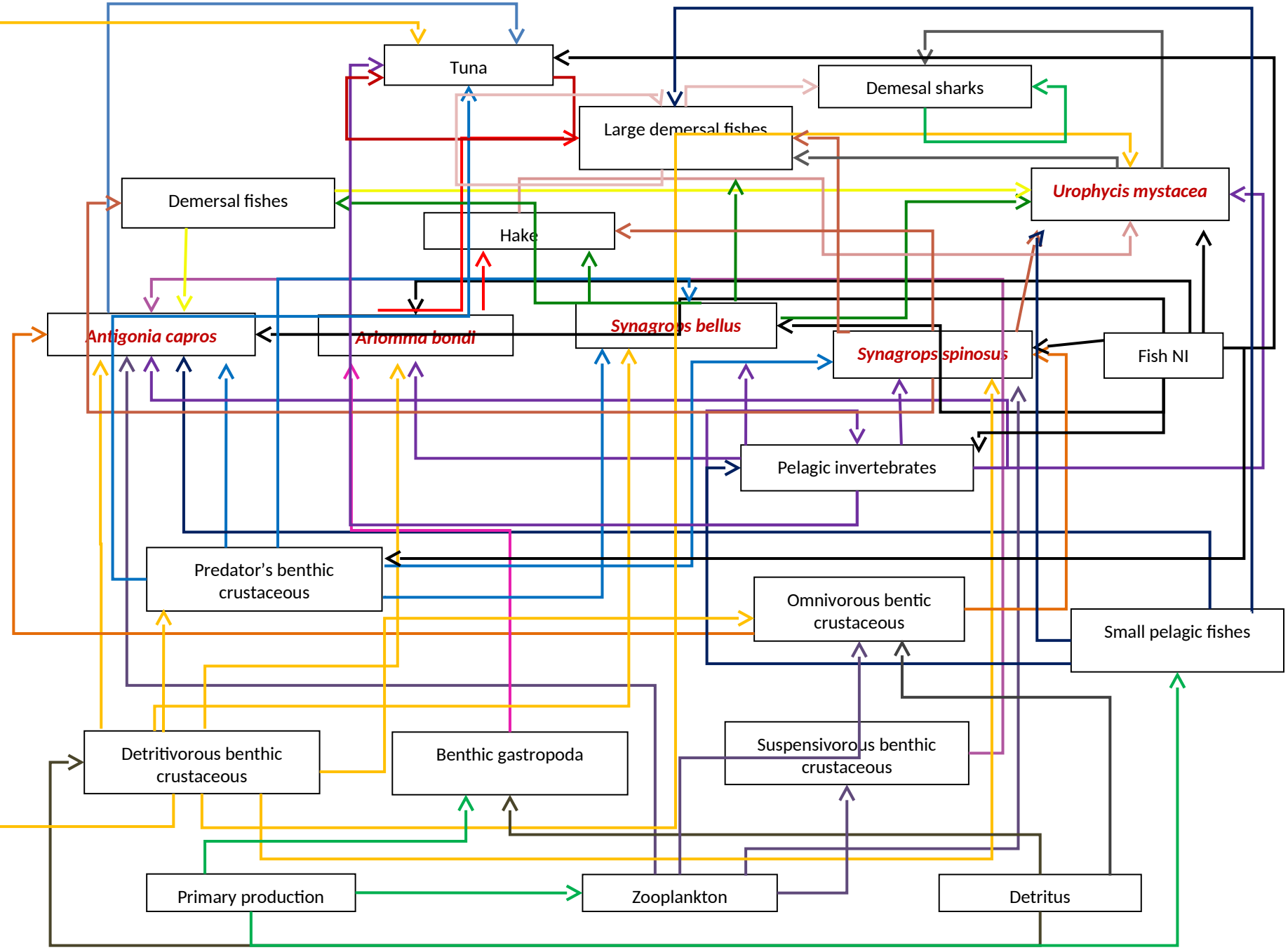


1331

Ariomma bondi
(Ariommatidae, Perciforme)

How is the ecosystem that these species live?

How important is each species in this ecosystem?


Tuna
Demesal sharks
Large demersal fishes
Demersal fishes
Urophycis mystacea
Hake
Antigonia capros
Ariomma bondi
Synagrops bellus
Synagrops spinosus
Fish NI
Pelagic invertebrates
Predator's benthic crustaceous
Omnivorous bentic crustaceous
Small pelagic fishes
Detritivorous benthic crustaceous
Benthic gastropoda
Suspensivorous benthic crustaceous
Primary production
Zooplankton
Detritus

37 Functional groups based on:
- Habitat
- Body size
- Feeding ecology

1-Blackfin goosefish
2-Cutlassfish
3-Sarrão
4-Tile fish
Hake
5-Juvenile
6-Adult
7-Searobin
8-Red porgy
9-Demersal Sharks
10-Sciaenidae
11-Bentho-pelagic fishes
12-Large bento-pelagic fishes
13-Antigonia capros
14-Ariomma bondi
15-Genypterus brasiliensis
16-Synagrops sp.
17-Urophycis mystacea
18-Ventrifossa macropogon
19-Squid
20-Large demersal fishes
21-Demersal fishes
22-Benthic fishes
23-Skates
24-Predator crustaceans
25-Predator mollusks
26-Another predator invertebrates
27-Omnivore invertebrates
28-Crab
29-Shrimp
30-Bivalve
31-Polychaetes
32-Another detritivore invertebrates
33-Infauna
34-Zooplankton
35-Phytoplankton
36-Marine snow
37-Detritus

Input data modified to balance the model, and the mainly output data. TL = Trophic level; Bf = Final biomass (t*km-2);P/B = Production/biomass ratio (Yr -1); Q/B = Consumption/Biomass ratio (Yr -1); EE = Ecotrophic Efficiency; P/Q = production/consumption ratio or Gross efficiency; F = Fishing mortality (yr−1); M2 = Predation mortality (yr−1); Mo = Other natural mortality (yr−1); F/Z = Exploitation rate; OI = Omnivory index; NE = Net efficiency; R/A =respiration/assimilation ratio; FD = Flow to detritus (t·km−2·yr−1).

Functional groups	TL	B_f	P/B	Q/B	EE	P/Q	F	M_2	M_0	F/Z	OI	NE	R/A	FD
Blackfin goosefish	4,708	0,115	0,4	2,3	0,572	0,174	0,059	0,170	0,171	0,149	0,276	0,217	0,783	0,072
Cutlassfish	4,652	0,063	0,91	3,1	0,972	0,294	0,029	0,836	0,026	0,033	1,721	0,367	0,633	0,041
Sarrão	4,179	0,016	0,8	4,8	0,362	0,167	-	0,290	0,510	-	0,752	0,208	0,792	0,023
Tile fish	4,640	0,021	0,74	2,5	0,329	0,296	0,062	0,182	0,497	0,083	2,172	0,370	0,630	0,021
Hake														
Juvenile	3,335	0,797	2,3	7,199	0,981	0,319	0,008	2,241	0,044	0,003	0,435	0,399	0,601	1,183
Adult	4,310	0,360	0,95	2,7	0,926	0,352	0,004	0,876	0,070	0,004	1,470	0,440	0,560	0,220
Searobin	3,161	0,016	0,38	2,5	0,527	0,152	0,197	0,003	0,180	0,518	0,135	0,190	0,810	0,011
Red porgy	3,400	0,009	0,89	3,5	0,950	0,254	0,637	0,208	0,045	0,716	0,377	0,318	0,682	0,007
Demersal Sharks	3,978	0,019	0,36	2,300	0,053	0,157	0,019	0,000	0,341	0,053	1,049	0,196	0,804	0,015
Sciaenidae	3,514	0,090	0,6	4,3	0,886	0,140	0,004	0,528	0,068	0,007	0,422	0,174	0,826	0,083
Bento-pelagic fishes	3,640	0,160	0,73	4,1	0,719	0,178	-	0,505	0,205	-	1,102	0,223	0,777	0,164
Large bento-pelagic fishes	3,847	0,477	0,3	2	0,991	0,150	-	0,278	0,003	-	0,128	0,188	0,813	0,192
Antigonia capros	3,159	0,203	0,61	5	0,312	0,122	0,003	0,158	0,420	0,005	0,116	0,153	0,848	0,289
Ariomma bondi	3,395	0,252	1,05	4,6	0,988	0,228	-	1,037	0,013	-	0,338	0,285	0,715	0,235
Genypterus brasiliensis	4,691	0,052	0,5	2,8	0,919	0,179	0,072	0,387	0,040	0,145	0,698	0,223	0,777	0,031
Synagrops sp.	3,855	0,204	1,11	5,6	0,985	0,198	-	1,090	0,000	-	0,636	0,248	0,752	0,226
Urophycis mystacea	4,797	0,075	0,7	2,5	0,994	0,280	0,033	0,654	0,004	0,047	0,466	0,350	0,650	0,038
Ventrifossa macropogon	3,121	0,003	0,62	2,9	0,900	0,214	0,138	0,419	0,062	0,223	0,102	0,267	0,733	0,002
Squid	3,967	0,256	6,7	36,50	0,981	0,184	0,003	6,523	0,124	-	1,725	0,229	0,771	1,902
Large demersal fishes	3,497	0,038	1,08	4,20	0,855	0,257	0,227	0,697	0,156	0,210	0,223	0,321	0,679	0,038
Demersal fishes	3,478	0,447	1,14	3,90	0,981	0,292	0,004	1,114	0,022	0,004	0,571	0,365	0,635	0,358
Benthic fishes	3,484	0,280	1,3	5,175	0,736	0,251	0,025	0,932	0,343	0,019	0,310	0,314	0,686	0,386
Skates	3,438	0,051	0,14	4,4	0,115	0,032	0,016	-	0,124	0,115	0,379	0,040	0,960	0,052
Predator crustaceans	3,715	0,150	5	14,5	0,523	0,345	-	2,615	2,385	-	0,408	0,431	0,569	0,793
Predator mollusks	3,334	0,288	4,5	20	0,873	0,225	-	3,927	0,573	-	0,263	0,281	0,719	1,317
Another predator invertebrates	3,353	0,899	3,5	20	0,744	0,175	0,002	2,603	0,895	0,001	0,226	0,219	0,781	4,401
Omnivore invertebrates	3,000	4,495	12	12	0,990	1,000	0,002	11,878	0,120	-	0,477	1,250	-0,250	11,328
Crab	2,050	1,580	3,17	19	0,808	0,167	-	2,560	0,610	-	0,048	0,209	0,791	6,967
Shrimp	2,050	2,673	6	20	0,348	0,300	-	2,088	3,912	-	0,048	0,375	0,625	21,149
Bivalve	2,000	15,493	0,6	20,83	0,751	0,029	-	0,451	0,149	-	-	0,036	0,964	66,855
Polychaetes	2,850	7,709	3,5	20,83	0,688	0,168	-	2,408	1,092	-	0,727	0,210	0,790	40,538
Another detritivore invertebrates	2,000	4,215	4,5	12,2	0,429	0,369	-	1,930	2,570	-	-	0,461	0,539	21,117
Infauna	2,000	4,215	14,6	40	0,598	0,365	-	8,730	5,870	-	-	0,456	0,544	58,463
Zooplankton	2,000	3,6	104	248	0,092	0,419	-	9,530	94,470	-	-	0,524	0,476	518,652
Phytoplankton	1,000	15	182,96	-	-	-	-	-	182,960	-	-	-	-	2744,4
Marine snow	1,000	15	-	-	0,468	-	-	-	-	-	-	-	-	-
Detritus	1,000	30	-	-	0,418	-	-	-	-	-	0,086	-	-	-

Diet matrix adjusted to balance the model

ECOPATH input data

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
1 Blackfin goosefish						0,02																												
2 Cutlassfish				0,03448		0,04993				0,00601																								
3 Sarrão				0,004					0,1																									
4 Tile fish					0,000	0,002			0,043																									
5 Hake juvenile	0,205			0,05113		0,05			0,02854	0,022							0,136		0,176															
6 Hake adult	0,05019			0,012					0,025	0,02							0,064		0,030															
7 Searobin				0,001																														
8 Red porgy		0,01																																
9 Demersal Sharks																																		
10 Sciaenidae								0,02006											0,005															
11 Bento-pelagic fishes			0,204	0,04948													0,010				0,035													
12 Large bento-pelagic fishes	0,28001	0,2	0,198	0,04948													0,010																	
13 *Antigonia capros*	0,02	0,04	0,01												0,050						0,0050		0,010											
14 *Ariomma bondi*	0,03	0,000	0,01			0,01									0,042	0,100			0,010		0,010	0,007	0,010											
15 *Genypterus brasiliensis*	0,000	0,020	0,015			0,01											0,020			0,01														
16 *Synagrops sp.*	0,106		0,005			0,01			0,00298	0,01682					0,100		0,020		0,010		0,020	0,015	0,030											
17 *Urophycis mystacea*	0,01201	0,02	0,01	0,01979		0,005									0,244																			
18 *Ventrifossa macropogon*															0,010																			
19 Squid	0,08004	0,015	0,152	0,09896	0,1	0,22008		0,052	0,19861	0,01402	0,04				0,102	0,152	0,508	0,001	0,05002		0,029													
20 Large demersal fishes															0,158					0,009			0,00999										
21 Demersal fishes	0,09		0,005												0,098	0,249	0,030				0,028	0,037		0,005	0,010									
22 Benthic fishes															0,013						0,075	0,010		0,010	0,016									
23 Skates																																		
24 Predator crustaceans						0,03801	0,04			0,050			0,004	0,01	0,040	0,0120	0,016				0,002	0,040	0,07992	0,100										
25 Predator mollusks	0,001	0,205						0,12036			0,001		0,001	0,26003									0,02098	0,11		0,010								
26 Another predator invertebrates												0,1																						
27 Omnivore invertebrates					0,100	0,020	0,005			0,10019	0,14	0,35	0,1																					
28 Crab			0,285	0,02969		0,03982	0,4	0,04012	0,17884	0,1802	0,18	0,003																						
29 Shrimp				0,02969	0,100	0,16407	0,5	0,30191	0,28877	0,14015		0,002	0,78	0																				
30 Bivalve						0,01801																												
31 Polychaetes			0,005				0,05	0,1003		0,19021	0,075	0,4																						
32 Another detritivore invertebrates			0,0375					0,15447			0,022		0,01																					
33 Infauna			0,02					0,20061		0,19021		0,07																						
34 Zooplankton					0,6		0,005			0,07		0,05	0,03																					
35 Marine snow																																		
36 Detritus																																		
37 Import	0,125	0,49	0,0435	0,62	0,1	0,343		0,01	0,1343		0,5425	0,025	0,075																					
Sum	1	1	1	1	1	1	1	1	1	1	1	1	1																					

	1	2	3	4
1 Blackfin goosefish				
2 Cutlassfish				0,03448
3 Sarrão				0,004
4 Tile fish				
5 Hake juvenile	0,205			0,05113
6 Hake adult	0,05019			0,012
7 Searobin				0,001
8 Red porgy		0,01		
9 Demersal Sharks				
10 Sciaenidae				
11 Bento-pelagic fishes			0,204	0,04948
12 Large bento-pelagic fishes	0,28001	0,2	0,198	0,04948
13 *Antigonia capros*	0,02	0,04	0,01	

Biomass of some groups

High Ecotrophic Efficiency → highly exploited

High Ecotrophic Efficiency → high level of predation

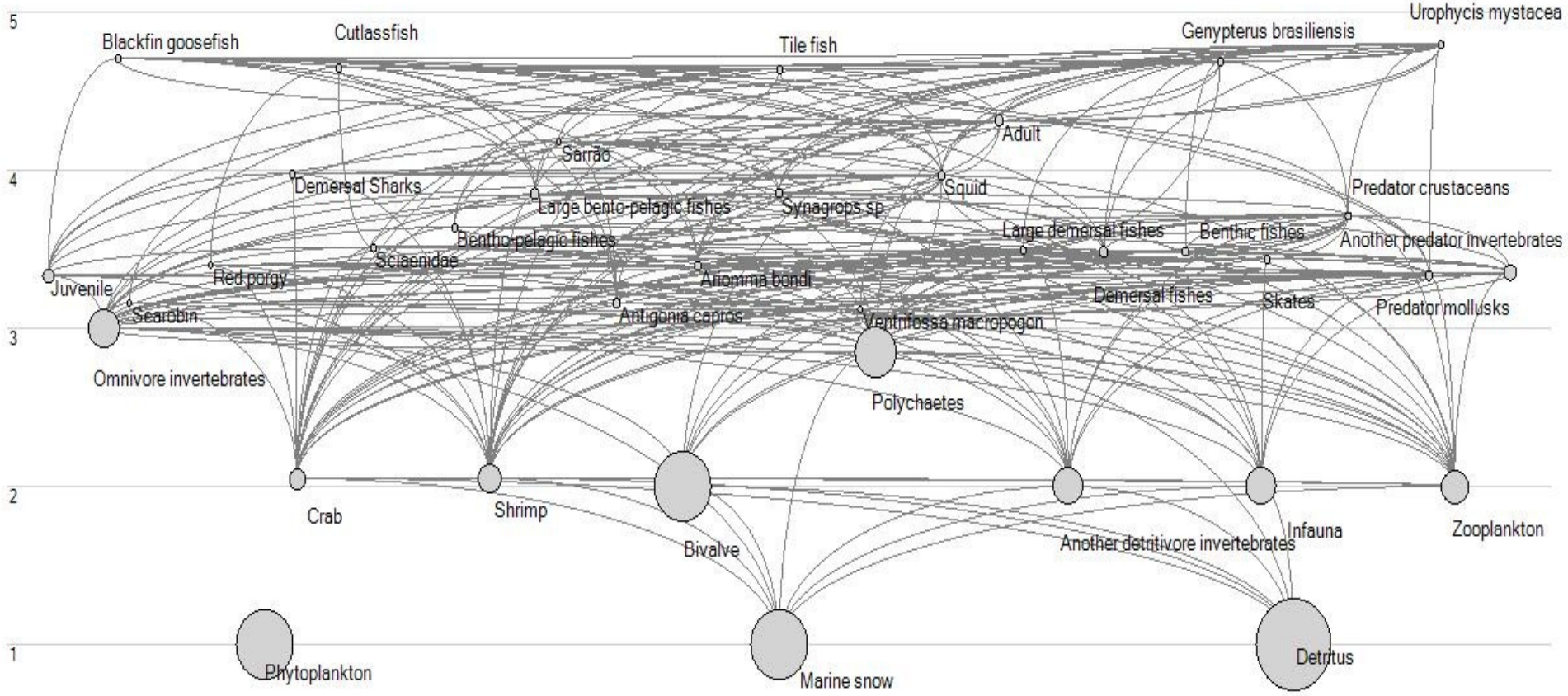
Cutlassfish → Heavily fished (also often dicarded), and pursued by the sport fishery

Genypterus brasiliensis
Urophycis mystacea
Squid
Crab

→ Highly consumed and fished in the system, being some of the most important resources in the demersal trawls

Demersal fishes,
Large demersal fishes
Large bentho-pelagic
Ariomma bondi
Synagrops sp.
Ventrifossa macropogon

→ Heavily consumed and important components in this system


5
4
3
2
1
Blackfin goosefish
Cutlassfish
Tile fish
Genypterus brasiliensis
Urophycis mystacea
Adult
Sarrão
Demersal Sharks
Squid
Large bento-pelagic fishes
Synagrops sp.
Predator crustaceans
Large demersal fishes
Benthic fishes
Another predator invertebrates
Bentho-pelagic fishes
Sciaenidae
Red porgy
Juvenile
Ariomma bondi
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Omnivore invertebrates
Polychaetes
Crab
Shrimp
Bivalve
Infauna
Another detritivore invertebrates
Zooplankton
Phytoplankton
Marine snow
Detritus

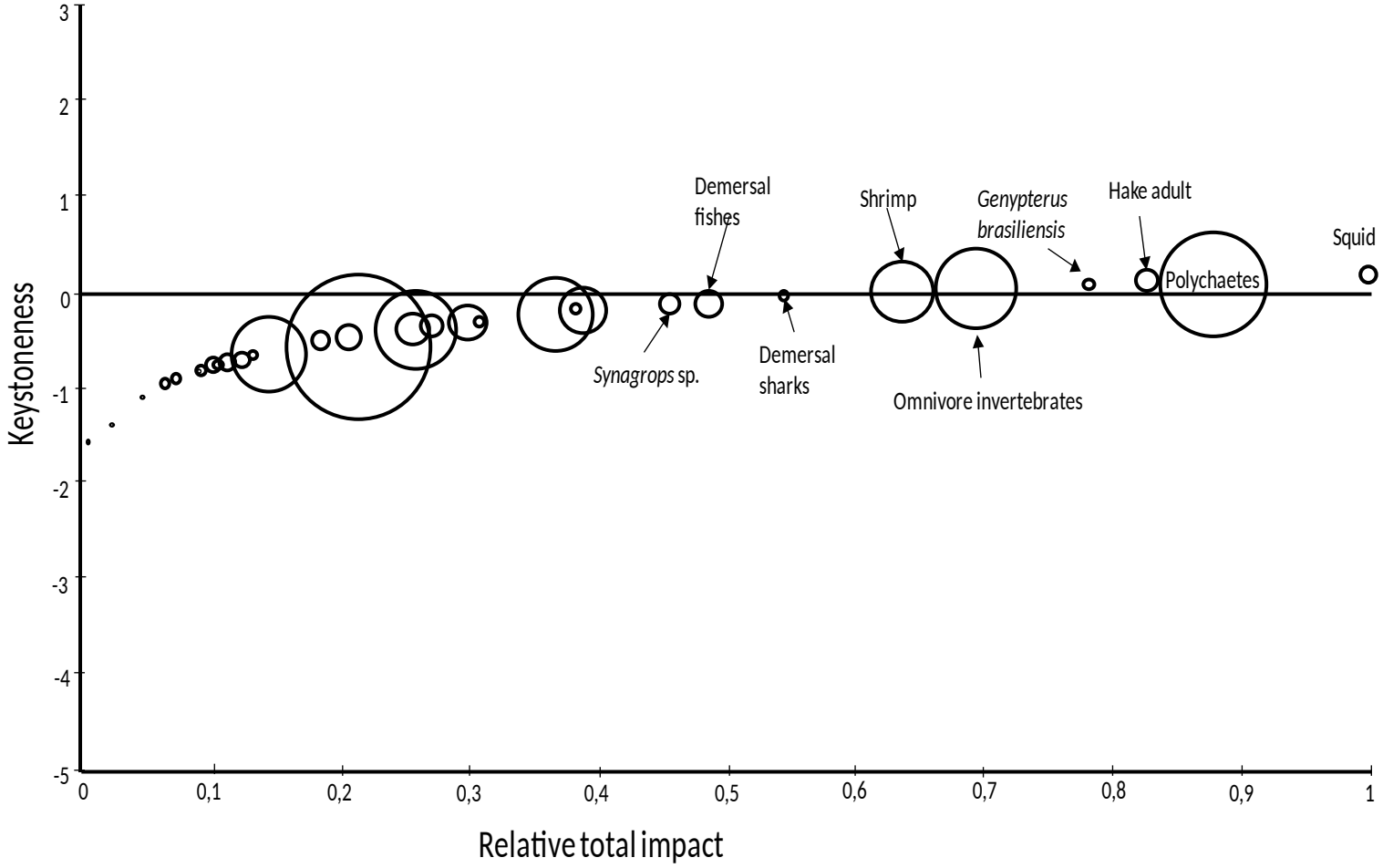
Ecosystem based on detritus

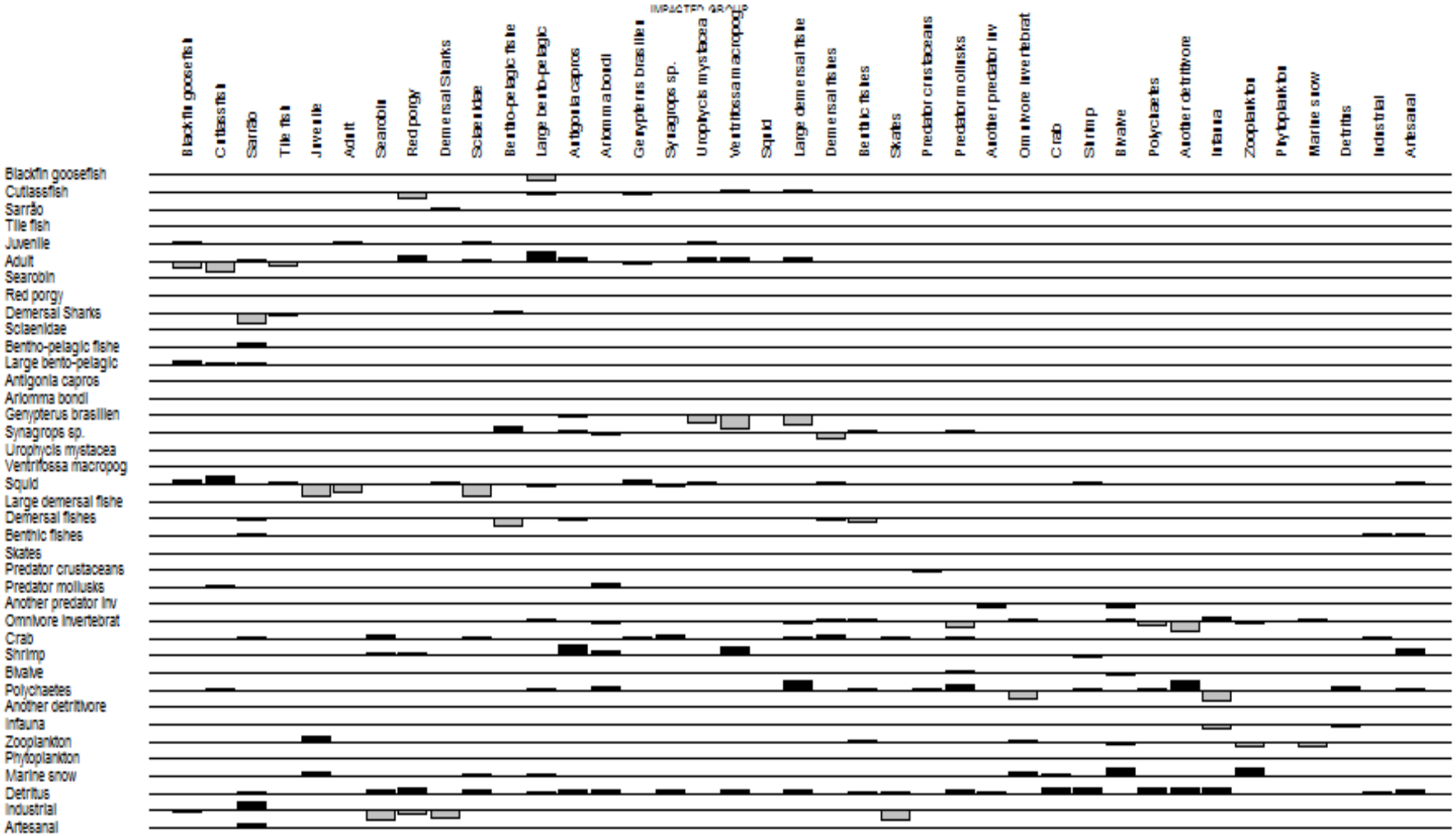
High trophic level of catches (3.5)

High value of conectance of the system (0.224)

High value system omnivory index (0.422)

High level of relative ascendancy (49.1%)


Keystoneness
Relative total impact
3
2
1
0
-1
-2
-3
-4
-5
0
0,1
0,2
0,3
0,4
0,5
0,6
0,7
0,8
0,9
1
Demersal fishes
Shrimp
Genypterus brasiliensis
Hake adult
Squid
Polychaetes
Synagrops sp.
Demersal sharks
Omnivore invertebrates


IMPACTED GROUP
Blackfin goosefish
Cutlassfish
Sarrão
Tile fish
Juvenile
Adult
Searobin
Red porgy
Demersal Sharks
Sciaenidae
Bentho-pelagic fishe
Large bento-pelagic
Antigonia capros
Ariomma bondi
Genypterus brasilien
Synagrops sp.
Urophycis mystacea
Ventrifossa macropog
Squid
Large demersal fishe
Demersal fishes
Benthic fishes
Skates
Predator crustaceans
Predator molluscs
Another predator inv
Omnivore invertebrat
Crab
Shrimp
Bivalve
Polychaetes
Another detritivore
Infauna
Zooplankton
Phytoplankton
Marine snow
Detritus
Industrial
Artesanal

Conclusions

Southeastern Brazilian Bight outer continental shelf and upper slope biological communities are supported prominently by detritus, with a high diversity of detritus-supported species and high omnivory. The demersal fishes are a key, highly connected link between the detritus-supported lower trophic levels and water column communities. This appears to be a system with a high capacity to recover from disturbances, but one that is still being depleted by uncontrolled fishing that has developed in the area over the last 20 years.

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Thank you